

Experiment 7

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Question 1: Find velocity in circular motion.

Formula: $v = \omega \times r$

Algorithm

1. Read inputs.
2. Compute $v = \omega \times r$.
3. Print result.
4. Stop.

C Program

```
#include <stdio.h>

int main()
{
    float angularVelocity, radius, velocity;

    printf("Enter angular velocity: ");
    scanf("%f", &angularVelocity);

    printf("Enter radius: ");
    scanf("%f", &radius);

    velocity = angularVelocity * radius;

    printf("Velocity = %.2f\n", velocity);

    return 0;
}
```

Test	Input	Expected	Actual	Result
1	2,5	10	10	Pass
2	4,3	12	12	Pass
3	1.5,8	12	12	Pass

Question 2: Find potential difference.

Formula: $V = W / Q$

Algorithm

1. Read inputs.
2. Compute V .
3. Print.
4. Stop.

C Program

```
#include <stdio.h>

int main()
{
```

```

float work, charge, potentialDifference;

printf("Enter work done: ");
scanf("%f", &work);

printf("Enter charge: ");
scanf("%f", &charge);

potentialDifference = work / charge;

printf("Potential Difference = %.2f\n", potentialDifference);

return 0;
}

```

Test	Input	Expected	Actual	Result
1	20,4	5	5	Pass
2	50,10	5	5	Pass
3	90,30	3	3	Pass

Question 3: Find horizontal component.

Formula: $H = M \times \cos(\theta)$

Algorithm

1. Read inputs.
2. Compute H.
3. Print.
4. Stop.

C Program

```

#include <stdio.h>
#include <math.h>

int main()
{
    float magnitude, angle, horizontal;

    printf("Enter magnitude: ");
    scanf("%f", &magnitude);

    printf("Enter angle (degrees): ");
    scanf("%f", &angle);

    horizontal = magnitude * cos(angle * 3.14159 / 180);

    printf("Horizontal Component = %.2f\n", horizontal);

    return 0;
}

```

Test	Input	Expected	Actual	Result
1	10,60	5.00	5.00	Pass
2	20,0	20.00	20.00	Pass
3	15,30	12.99	12.99	Pass